

PURPOSE

Time-development of the concrete secant modulus of elasticity, from cement class and mean 28-day strength.
Shared upstream of crack-width, deflection and bending-stiffness calculations.

1 Input Parameters

Symbol	Value	Unit	Description
s_{cem}	0.25	-	Cement class coefficient (0.20 R / 0.25 N / 0.38 S)
t	7	days	Concrete age at loading
f_{cm}	38	MPa	Mean compressive strength at 28 days (C30/37)
E_{cm}	32837	MPa	Secant modulus at 28 days (C30/37)

2 Calculation (all formulas from the shared library)

Concrete strength development coefficient at age t

$$\begin{aligned}\beta_{cc,t} &= e^{s_{cem} \cdot \left(1 - \sqrt{\frac{28}{t}}\right)} && \text{EN 1992-1-1:2004 §3.1.2(6), Eq. (3.2)} \\ &= e^{0.25 \cdot \left(1 - \sqrt{\frac{28}{7}}\right)} \\ &= \mathbf{0.7788}\end{aligned}$$

Mean concrete compressive strength at age t

$$\begin{aligned}f_{cm,t} &= \beta_{cc,t} \cdot f_{cm} && \text{EN 1992-1-1:2004 §3.1.2(6), Eq. (3.1)} \\ &= 0.7788 \cdot 38 \\ &= \mathbf{29.59 \text{ MPa}}\end{aligned}$$

Secant modulus of elasticity at age t

$$\begin{aligned}E_{cm,t} &= E_{cm} \cdot \left(\frac{f_{cm,t}}{f_{cm}}\right)^{0.3} && \text{EN 1992-1-1:2004 §3.1.3(3), Eq. (3.5)} \\ &= 32837 \cdot \left(\frac{29.59}{38}\right)^{0.3} \\ &= \mathbf{30464 \text{ MPa}}\end{aligned}$$

3 Verification

OK

Modulus at early age does not exceed the 28-day value

$$E_{cm,t} = 30464 \text{ MPa} \leq E_{cm} = 32837 \text{ MPa}$$